NATIONAL MATHEMATICS CONTEST

A-LEVEL PAPER 1.

SATURDAY 6TH APRIL, 2019 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.

(b) There are **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) Indicate your names, birth date, gender, class, school and district ***on all*** *your answer sheets.*

(d) Answer scripts of only registered students should be returned to the contest coordinator, latest **Monday 8th April, 2019**.

(e) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 27*th July,* 2019 *at NOON at Makerere University, Kampala.*

(f) National Mathematics Contest 2019 *Paper* 2 *will be done on* 29*th June,* 2019 *at various centres.*

SECTION A : 5 marks each

1. Find the root of the equation $2 + \log\sqrt{1+x} + 3\log\sqrt{1-x} = \log\sqrt{1-x^2}$.

2. Let ABC be a triangle. Let P be the point on line BC such that B is between P and C, and $BP = BA$. Similarly, let Q be the point on line BC such that C is between Q and B, and $CQ = CA$. If R is the second intersection of the circumcircles of $\triangle ACP$ and $\triangle ABQ$, prove that $\triangle PQR$ is isosceles.

3. A deck contains 52 cards of 4 different suits. Vanya is told the number of cards in each suit. He picks a card from the deck, guesses its suit, and sets it aside; he repeats until the deck is exhausted. Show that if Vanya always guesses a suit having no fewer remaining cards than any other suit, he will guess correctly at least 13 times.

4. Find, with proof, all real number solutions to the following:

$$\left(a^2+1\right)\left(b^2+1\right) = (ab+1)(a+b).$$

5. Solve the equation

$$\sqrt{-3x^2+18x+37} + \sqrt{-5x^2+30x-41} = \sqrt{x^2-6x+109}$$

6. For positive real numbers a, b, c such that $abc \leq 1$, prove that

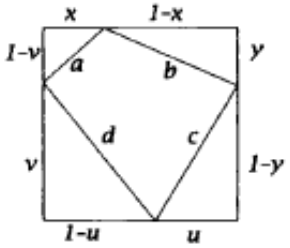
$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \;\geq\; a+b+c.$$

7. Determine the value of the expression

$$1+2-3+4+5-6+7+8-9+10+11-12+\cdots+94+95-96+97+98-99.$$

8. A quadrilateral has one vertex on each side of a square of side-length 1. Show that the lengths a, b, c and d of the sides of the quadrilateral satisfy the inequalities

$$2 \leq a^2+b^2+c^2+d^2 \leq 4.$$



9. Without using a calculator, evaluate

$$\sqrt[3]{2\sqrt{13}+5}-\sqrt[3]{2\sqrt{13}-5}.$$

10. Let a, b and c be real numbers such that $a+b+c=5$ and $\dfrac{1}{b+c}+\dfrac{1}{c+a}+\dfrac{1}{a+b}=6$. Evaluate the value of $\dfrac{a}{b+c}+\dfrac{b}{c+a}+\dfrac{c}{a+b}$.

SECTION B : 10 marks each

11. Determine x, if

$$\begin{aligned} 4^{\frac{x}{y}+\frac{y}{x}} &= 32 \\ \log_3(x-y)+\log_3(x+y) &= 1 \end{aligned}$$

12. Let $ABCD$ be a parallelogram with $AC/BD=k$. The bisectors of the angles formed by AC and BD intersect the sides of $ABCD$ at $K,\ L,\ M,\ N$. Prove that the ratio of the areas of $KLMN$ and $ABCD$ is $\dfrac{2k}{(k+1)^2}$.

13. (a) Find the number of subsets of $\{1, 2, \cdots, 10\}$ that contain its own size. For example, the set $(1, 3, 6)$ has 3 elements and contains 3.

    (b) Every vertex of the unit squares on an $n \times m$ grid is coloured either blue, green, or red, such that all the vertices on the boundary of the board are coloured red. We say that a unit square on the board is properly coloured if exactly one pair of adjacent vertices of the square are the same colour. Show that the number of properly coloured squares is even.

14. Let $x, y, z \geq 0$ be such that $x+y+z=3$. Prove that

$$\frac{x^3}{y^3+8}+\frac{y^3}{z^3+8}+\frac{z^3}{x^3+8} \geq \frac{1}{9}+\frac{2}{27}\cdot(xy+yz+zx)$$

When does equality hold?

15. Let a positive integer k be given. Prove that there are infinitely many pairs of integers (a, b) with $|a|>1$ and $|b|>1$ such that a^2+b^2+k is divisible by $ab+a+b$.

NATIONAL MATHEMATICS CONTEST
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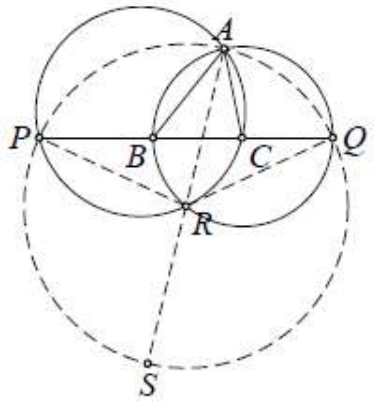
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SECTION A : 5 marks each

1. The equation is defined for $1+x>0, 1-x>0$ and $1-x^2>0$. The third one follows from the first two, so we can exclude it. What remains is $x \in (-1,1)$

$$\begin{aligned}
2+\log\sqrt{1+x}+3\log\sqrt{1-x} &= \log\sqrt{1-x^2} \\
2+\log\sqrt{1+x}+3\log\sqrt{1-x} &= \log\sqrt{1-x}+\log\sqrt{1+x} \\
2+2\log\sqrt{1-x} &= 0 \\
\log\sqrt{1-x} &= -1 \\
\sqrt{1-x} &= 10^{-1}=\frac{1}{10} \\
1-x &= \frac{1}{100} \\
x &= \frac{99}{100}
\end{aligned}$$

2.



Let S be the intersection of AR with the circumcircle of $\triangle PAQ$. Then:

$$\begin{aligned} \angle PSA &= \angle AQP = \frac{1}{2}\angle ACP = \frac{1}{2}\angle ARP \quad \Rightarrow \\ \angle PSA &= \angle SPR \quad \Rightarrow \\ RP &= RS \end{aligned}$$

Similarly $RQ = RS$. Hence $RP = RQ$.

3. Let M denote the maximum number of cards remaining in any single suit. As Vanya proceeds, M will only decrease if the current card is in a suit with M cards remaining, and no other suit has M cards remaining. In this case, however, Vanya will correctly guess that suit. Therefore, Vanya will guess correctly every time M decreases. Since $M \geq \dfrac{52}{4} = 13$ initially and it is 0 by the end, Vanya will be correct at least 13 times.

4. We have

$$\begin{aligned} & 2(a^2+1)(b^2+1) - 2(ab+1)(a+b) \\ = \; & 2(a^2b^2 + a^2 + b^2 + 1 - a^2b - ab^2 - a - b) \\ = \; & (a^2b^2 - 2a^2b + a^2) + (a^2b^2 - 2ab^2 + b^2) + (a^2 - 2a + 1) + (b^2 - 2b + 1) \\ = \; & a^2(b-1)^2 + b^2(a-1)^2 + (a-1)^2 + (b-1)^2 \\ = \; & (b^2+1)(a-1)^2 + (a^2+1)(b-1)^2 \end{aligned}$$

which is positive unless $a = 1$ and $b = 1$, which forms a solution. Therefore, the only solution is $a = b = 1$.

Remark: Alternatively, note that by Cauchy-Schwarz inequality, we have

$$(a^2+1)(b^2+1) \geq (ab+1)^2 \quad \text{and} \quad (a^2+1)(1+b^2) \geq (a+b)^2$$

and multiplying together yields $[(a^2+1)(b^2+1)]^2 \geq [(ab+1)(a+b)]^2$, so we can find all the solutions to the equation by consider the equality cases in the above inequalities.

5. After setting $t = (x-3)^2$, the equation reduces to

$$\sqrt{64 - 3t} + \sqrt{4 - 5t} = \sqrt{t + 100}.$$

Then $t = 0$ is one solution, and the left-hand side is decreasing in t while the right-hand side is increasing in t. Therefore, $t = 0$ (and hence $x = 3$) is the unique solution.

6. Using the weighted AM-GM, we have

$$\frac{2a}{b}+\frac{b}{c} \geq 3\cdot\sqrt[3]{\frac{a^2}{bc}} \geq 3a.$$

Similarly,

$$\frac{2b}{c}+\frac{c}{a} \geq 3\cdot\sqrt[3]{\frac{b^2}{ca}} \geq 3b$$
$$\frac{2c}{a}+\frac{a}{b} \geq 3\cdot\sqrt[3]{\frac{c^2}{ab}} \geq 3c$$

Adding all the three inequalities,

$$\begin{aligned}\frac{2a}{b}+\frac{b}{c}+\frac{2b}{c}+\frac{c}{a}+\frac{2c}{a}+\frac{a}{b} &\geq 3a+3b+3c\\ 3\left(\frac{a}{b}+\frac{b}{c}+\frac{c}{a}\right) &\geq 3(a+b+c)\\ \frac{a}{b}+\frac{b}{c}+\frac{c}{a} &\geq a+b+c\end{aligned}$$

giving the result.

Alternatively,

Replacing a, b, c by ta, tb, tc with $t=\dfrac{1}{\sqrt[3]{abc}}$ preserves the LHS while increases the RHS and makes at $at\cdot bt\cdot ct = abct^3 = 1$. Hence we may assume without loss of generosity that $abc = 1$.

Then there exists positive real numbers x, y, z such that

$$a=\frac{y}{x},\ b=\frac{z}{y},\ c=\frac{x}{z}$$

The rearrangement of the inequality gives

$$x^3+y^3+z^3 \geq x^2y+y^2z+z^2x$$

Thus

$$\frac{a}{b}+\frac{b}{c}+\frac{c}{a}=\frac{x^3+y^3+z^3}{xyz}\geq\frac{x^2y+y^2z+z^2x}{xyz}=a+b+c$$

as desired

7. Removing the initial 0, the remaining 99 terms can be written in groups of the form

$$(3k-2)+(3k-1)-3k \text{ for each } k \text{ from 1 to 33.}$$

The expression $(3k-2)+(3k-1)-3k$ simplifies to $3k-3$. Therefore, the given sum equals

$$0+3+6+\cdots+93+96$$

Since k ran from 1 to 33, then this sum includes 33 terms and so equals

$$\frac{33}{2}(0+96) = 33(48) = 33(50) - 33(2) = 1650 - 66 = 1584$$

8. First note that

$$a^2+b^2+c^2+d^2 = x^2+(1-x)^2+y^2+(1-y)^2+u^2+(1-u)^2+v^2+(1-v)^2.$$

Now consider

$$x^2+(1-x)^2 = 2\left\{\left(x-\frac{1}{2}\right)^2+\frac{1}{4}\right\}.$$

Since $0 \le x \le 1$, it follows easily that $\frac{1}{2} \le x^2+(1-x)^2 \le 1$, and similarly with x replaced by y, u, v. Adding the four inequalities yields the desired result.

9. Let $A = \sqrt[3]{2\sqrt{13}+5}$ and $B = \sqrt[3]{2\sqrt{13}-5}$. Then we wish to solve for $A-B$. Notice that

$$\begin{aligned} A^3-B^3 &= \left(2\sqrt{13}+5\right)-\left(2\sqrt{13}-5\right) = 10 \\ AB &= \sqrt[3]{\left(2\sqrt{13}+5\right)\left(2\sqrt{13}-5\right)} = 3 \end{aligned}$$

such that

$$\begin{aligned} (A-B)^3 &= A^3-3A^2B+3AB^2-B^3 \\ &= A^3-B^3-3AB(A-B) \\ &= 10-3(3)(A-B) \\ &= 10-9(A-B) \\ (A-B)^3+9(A-B)-10 &= 0 \\ [(A-B)-1]\left[(A-B)^2+(A-B)+10\right] &= 0 \end{aligned}$$

since $(A-B)$ is real, $A-B=1$.

10. Note that

$$\begin{aligned} \frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} &= \frac{5-(b+c)}{b+c}+\frac{5-(c+a)}{c+a}+\frac{5-(a+b)}{a+b} \\ &= \frac{5}{b+c}+\frac{5}{c+a}+\frac{5}{a+b}-3 \\ &= 5(6)-3 \\ &= 27 \end{aligned}$$

SECTION B : 10 marks each

11. Checking if the system is defined for two variables is a hard task, so we shall find the eventual solutions to the system and check directly if the system is defined for them. We shall only write

$$\begin{aligned} x+y &> 0 \\ x-y &> 0 \end{aligned}$$

for now.

$$\begin{aligned} \frac{x}{y}+\frac{y}{x} &= \log_4 32 \\ \log_3(x^2-y^2) &= 1 \end{aligned}$$

$$\begin{aligned} \frac{x}{y}+\frac{y}{x} &= \frac{1}{2}\log_2 32 \\ x^2-y^2 &= 3 \end{aligned}$$

$$\begin{aligned} \frac{x^2+y^2}{xy} &= \frac{5}{2} \\ x^2-y^2 &= 3 \end{aligned}$$

$$\begin{aligned} x^2+y^2 &= \frac{5}{2}xy \\ x^2-y^2 &= 3 \end{aligned}$$

$$\begin{aligned} x^2-2xy+y^2 &= \frac{1}{2}xy \\ (x-y)(x+y) &= 3 \end{aligned}$$

$$\begin{aligned} (x-y)^2 &= \frac{xy}{2} \\ (x-y)(x+y) &= 3 \end{aligned}$$

We divide the first with the square of the second:

$$\frac{1}{(x+y)^2}=\frac{xy}{18} \quad \text{or} \quad (x+y)^2=\frac{18}{xy}.$$

We get the system

$$\begin{aligned} x^2+2xy+y^2 &= \frac{18}{xy} \\ x^2-2xy+y^2 &= \frac{xy}{2} \end{aligned}$$

we now subtract them

$$\begin{aligned} 4xy &= \frac{18}{xy} - \frac{xy}{2} \\ \frac{9}{2}xy &= \frac{18}{xy} \\ (xy)^2 &= 4 \\ xy &= \pm 2 \end{aligned}$$

But it can't be negative, because $2(x-y)^2 = xy$, therefore $xy = 2$. Substituting back into the system, we get

$$\begin{aligned} (x-y)^2 &= x^2 - 2xy + y^2 = x^2 + y^2 - 4 = 1 \\ x^2 - y^2 &= 3 \end{aligned}$$

$$\begin{aligned} x^2 + y^2 &= 5 \\ x^2 - y^2 &= 3 \end{aligned}$$

we subtract them: $2x^2 = 8$, so $x_1 = 2$ and $x_2 = -2$. $y_1 = \dfrac{2}{x_1} = 1$ and $y_2 = \dfrac{2}{x_2} = -1$. We now have to check if they are solutions. $x_1 + y_1 = 3 > 0$, $x_1 - y_1 = 1 > 0$, so $(2, 1)$ is a solution to the system. $x_2 + y_2 = -3 < 0$, so $(-2, -1)$ is not a solution. The final solution is $x = 2$.

12. Suppose K, L, M, N are on AB, BC, CD, DA, respectively. Let $P = AC \cap BD$. Since $ABCD$ is a parallelogram, $AP = PC$ and $BP = PD$. For any polygon $X_1X_2\cdots X_n$, we denote its area by $[X_1X_2\cdots X_n]$.

By the angle bisector theorem, $AK/KB = AP/PB = AC/BD = k$. Similarly, $AN/ND = k$. Hence, $AK = AB \models AN/AD = \dfrac{k}{k+1}$. Consequently, KN is parallel to BD. Furthermore,

$$[AKN] = \frac{k^2}{(k+1)^2}[ABD] = \frac{k^2}{2(k+1)^2}[ABCD].$$

Similarly,

$$\begin{aligned} [BKL] &= \frac{1}{2(k+1)^2}[ABCD], \\ [CLM] &= \frac{k^2}{2(k+1)^2}[ABCD], \\ [DMN] &= \frac{1}{2(k+1)^2}[ABCD]. \end{aligned}$$

Hence,

$$\begin{aligned} [KLMN] &= [ABCD] - [AKN] - [BKL] - [CLM] - [DMN] \\ &= \left[1 - \frac{k^2}{(k+1)^2} - \frac{1}{(k+1)^2}\right][ABCD] \\ &= \frac{2k}{(k+1)^2}[ABCD], \end{aligned}$$

as desired

13. (a) For each $k \in \{1, 2, \cdots, 10\}$, we want to find how many elements there are of size k. Note that $k \neq 0$ since $0 \notin \{1, 2, \cdots, 10\}$. Every subset of size k must contain k. The other $k-1$ elements can be anything. Therefore, there are $\binom{9}{k-1}$ subsets of size k that contains k. Therefore, the number of subsets that contain its own size is

$$\sum_{k=1}^{10} \binom{9}{k-1} = \sum_{k=0}^{9} \binom{9}{k} = 2^9 = 512$$

(b) We will call an edge of a square good if both its endpoints are of the same colour, and bad otherwise. Any square can have either no good edges (call such squares type A), one good edge (type B), two good edges (type C), or four good edges (type D). Note that it cannot have three good edges.

Let the number of squares of type A, B, C, D be a, b, c, d respectively. For each square, count the number of good edges, and add the numbers. We must get an even number, since each edge not on the boundary is counted twice, and all $2(m+n)$ edges on the boundary are good. On the other hand, this number is $b + 2c + 4d$. Hence b is even and the result follows.

Comments: We are counting in two different ways the number of (square, good edge) pairs. This kind of argument is very important in combinatorics problems!

14. Since $y \geq 0$, the AM-GM inequality implies

$$\frac{x^3}{y^3+8} + \frac{y+2}{27} + \frac{y^2-2y+4}{27} \quad \geq \quad 3 \cdot \sqrt[3]{\frac{x^3}{27^2}} = \frac{x}{3}$$

Similarly,

$$\begin{aligned} \frac{y^3}{z^3+8} + \frac{z+2}{27} + \frac{z^2-2z+4}{27} &\geq \frac{y}{3} \quad \text{and} \\ \frac{z^3}{x^3+8} + \frac{x+2}{27} + \frac{x^2-2x+4}{27} &\geq \frac{z}{3} \end{aligned}$$

Adding all the three inequalities, we have

$$\begin{aligned} \frac{x^3}{y^3+8} + \frac{y^3}{z^3+8} + \frac{z^3}{x^3+8} &\geq \frac{x+y+z}{3} + \frac{y+z+x}{27} - \frac{6}{9} - \frac{y^2+z^2+x^2}{27} \\ &= \frac{4}{9} - \frac{(x+y+z)^2 - 2xy - 2yz - 2zx}{27} \\ &= \frac{1}{9} + \frac{2}{27} \cdot (xy + yz + zx) \end{aligned}$$

For equality to hold, we must have $\frac{y+2}{27} = \frac{y^2-2y+4}{27} \Rightarrow y^2 - 3y + 2 = 0$, so y equals 1 or 2. The same holds for z and x. Since $x + y + z = 3$, the only possibility is $x = y = z = 1$, and it is easy to check that equality does indeed hold in this case.

15. By inspection, we see that if $(a, b) = (0, 1)$, then $a^2 + b^2 + k$ is divisible by $ab + a + b$.

$$\frac{a^2+b^2+k}{ab+a+b} \quad = \quad k+1 \tag{1.1}$$

Let us rearrange this as a quadratic in a:

$$a^2 - (k+1)(b+1)a + (b-1)(b-k) \quad = \quad 0.$$

As a quadratic in a, the sum of the roots is $(k+1)(b+1)$. Hence, if (a,b) is a solution to (1.1), then so is $((k+1)(b+1)-a,b)$, and hence by symmetry of (1.1), so is $(b,(k+1)(b+1)-a)$.

Define a sequence (a_n) as follows: $a_1=0, a_2=1$, and

$$a_n=(k+1)\left(a_{n-1}+1\right)-a_{n-2}$$

for all $n\geq 2$. Then by the above reasoning, $(a,b)=(a_n,a_{n+1})$ is a solution of (1.1) for all $n\geq 0$. Furthermore, the sequence (a_n) is increasing, giving an infinite number of positive integer solutions.